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ZentroMart Voice Support Agent

LLM + retrieval-augmented generation + verified tools + Gnani speech | Consulting & Analytics Club, IIT Guwahati x Gnani.ai

Executive summary

I built a customer support agent for the synthetic retailer ZentroMart. The agent answers from company documents, remembers the conversation, and checks who the caller is before it shares any order details. When a case needs a person, I hand it over by creating a ticket. I then connected the same agent to Gnani speech-to-text and text-to-speech, so a customer can speak and hear a reply.

I built the work in four steps and tested each one on its own. The key result is that adding retrieval from company documents raised decision accuracy from **9.6%** to **82.7%** on the 52 practice questions. I also ran the Groq-hosted language model and the Gnani voice services with real audio. The results below are the actual numbers from my full evaluation run.

82.7%	47 / 47	100%	17 / 17
Decision accuracy with RAG (baseline: 9.6%)	Questions with gold evidence retrieved at k = 6	Tool-call pass rate across 96 conversation turns	Local tests passed

Solution architecture

Layer	What I implemented	How I controlled it
Step 1: baseline	A prompt-only language model called through an OpenAI-compatible API. In my final run I used Groq with the openai/gpt-oss-120b model.	I gave it no company documents, no memory and no tools, so anything it did not know stayed unknown. This is my comparison point.
Step 2: retrieval (RAG)	I split the knowledge base into sections and added the product catalogue rows. Search combines word matching with sentence-transformer embeddings.	Each section keeps its ID, document type, effective date, status and source file, so every answer can be traced to evidence.
Step 3: conversation	The agent keeps session history and uses the supplied lookup_order and create_ticket tools, with fixed rules for verification and safety.	Order details reach the model only after the tool confirms the caller's phone suffix or delivery pincode.
Step 4: voice	Gnani speech-to-text turns the caller's audio into text. The same agent replies, and Gnani text-to-speech turns the reply into a WAV file.	I use no other speech provider. Each turn saves the transcript, reply, decision and retrieved evidence.

Design priorities I set

- **Policy order:** current policy first, then the agent guide, help pages, FAQ and marketing. Old policy sections stay marked as superseded. The agent uses them only when a question needs the earlier rules, such as an electronics delivery before 1 August 2026.
- **No false promises:** the supplied tools cannot cancel, refund, return or replace an order, so my agent never says it has done any of these. It describes the policy and offers a handover.
- **Honest tickets:** I report a ticket as created only when create_ticket returns created=True.
- **Privacy:** verification values, phone numbers, email, customer ID and delivery pincode are kept out of the model context and out of saved tool traces.

How the agent works

Finding the right evidence. When a customer asks something, the agent searches the knowledge base and passes the six best sections to the language model. I use top-six because my offline test showed it found every gold evidence label on the practice questions. I added small score boosts for common topics such as refunds, returns, warranty, cancellation, delivery and safety, so the governing policy is not pushed down by a similar product page.

Conversation and verification. The agent starts by saying it is an AI assistant. For order questions it asks for the order ID and either the last four digits of the phone number or the delivery pincode. The caller gets one retry. Until the tool confirms the details, the model never sees the order, so it cannot leak it.

Handover. If the customer asks for a person, or the case is beyond the agent (for example a refund that is overdue, or a high-value damaged item), the agent creates a ticket and shares the reference number. It does not approve replacements or promise outcomes.

Voice. The voice layer is a thin wrapper. Audio goes to Gnani speech-to-text, the transcript goes to the same agent, and the reply goes to Gnani text-to-speech. Because voice reuses the text agent, every control above also applies to voice.

Evaluation and validation

I measured the prompt-only baseline against the retrieval agent on all 52 practice questions, and I ran all 23 supplied conversations (96 turns) through the full agent. These figures come from my full run of the evaluation script.

Metric (52 questions)	Step 1: prompt only	Step 2: RAG
Decision accuracy	9.6%	82.7%
Required facts pass rate	15.4%	75.0%
Forbidden-claim pass rate	100.0%	100.0%
Combined pass rate	7.7%	61.5%
Evidence recall on answerable questions	0.0%	97.9%

The forbidden-claim score is 100% for both steps. For the baseline this is partly because it says very little and often asks for more information, so I do not treat it as a sign of quality.

Metric (23 conversations, 96 turns)	Result
Decision accuracy	78.1%
Required facts pass rate	85.4%
Forbidden-claim pass rate	100.0%
Tool-call pass rate	100.0%
Combined pass rate	75.0%

Check	Result	What it tells me
Supplied dataset validator	Passed	Tables, documents, keys, order arithmetic, evidence labels and verification rules are consistent.
Retrieval diagnostic (k = 6)	47/47 questions; 90/90 labels	Every question with gold evidence had at least one gold section in the top six, and all 90 labels were found. This is a development-set check, not proof for unseen questions.
Local tests	17 passed	They cover verification secrecy, one retry, ticket creation only on success, privacy refusal, overdue refund handover, historical policy retrieval, date parsing, the baseline having no tools, and both Gnani output styles.

Failure analysis and what I improved

My first retrieval version used five results and sometimes ranked similar product pages above the policy that actually answered the question. I fixed this step by step and checked the results after each change.

- **Intent boosts.** I added small score boosts for refunds, returns, warranty, cancellation, delivery and safety topics, without ignoring how well the text matches the question.
- **Catalogue cap.** Product rows are limited so that many near-identical SKUs cannot crowd out policy sections. An exact SKU mention still gets a strong match.
- **Festive-sale rule.** For a Hindi question about returning a canvas tote bag, the festive sale section (RET-8) first appeared at rank 3, behind two product rows. After I added a rule for bag and tote return questions, it moved to rank 1.
- **Opened beauty items.** For a question about an opened lipstick, the condition-of-returned-items section (RET-4) ranks second, directly after the product row, which is the order I wanted.
- **Historical policy.** The superseded returns policy is indexed with its status and is boosted only when a question points to a delivery before 1 August 2026.
- **Date handling.** Business-day arithmetic uses the fixed date of 20 October 2026, skips Sundays and listed holidays, and counts return windows in calendar days.
- **Top-k of six.** I raised it from five after offline measurement showed all 90 gold labels were found.

What still goes wrong. In the full run, about 17% of the 52 answers had the wrong decision (answer, clarify, refuse or escalate), and a quarter missed at least one required fact. In my smoke test, the agent refused a midnight-delivery question that should have been answered, and in another case it gave the right claim steps but left out the 12-month Assured period. These are the cases I would review first.

Live testing: what I ran

Area	What I did	What I observed
Language model	Connected Groq (openai/gpt-oss-120b) through the OpenAI-compatible interface and ran the full evaluation: 52 questions and 96 conversation turns.	The connection test returned a valid reply, and the full evaluation completed. A typed test with a damaged-product complaint produced an AI introduction and a request for order ID and verification details before any order help.
Gnani voice	Recorded three short audio files for one conversation (PC002) and ran them through Gnani speech-to-text, the agent, and Gnani text-to-speech.	Each turn produced a transcript, a text reply and a reply WAV file. The first turn about a delayed order was transcribed correctly and the agent asked for the order ID and verification details. The second, about scheduling a midnight delivery, led to a handover ticket.

Known limitations

- **Limited voice testing.** I tested voice with three short recordings from one scenario. I have not tested a wider range of accents, background noise, speaking speeds or longer calls, and I have not yet measured speech accuracy or response time.
- **Phrase-based scoring.** The automatic evaluator checks for expected phrases and facts. It can mark a correct answer wrong when the wording or a date format differs, so the real quality may be higher than some scores suggest. I read the failures by hand rather than trusting the score alone.
- **Development data only.** The 47/47 retrieval result and my tuning come from the practice questions. They do not prove the same accuracy on unseen questions.
- **Mixed-language review.** Hindi and Hinglish answers need a deeper manual review than I have done so far.
- **Limited actions.** The supplied tools can look up orders and create tickets only. The agent cannot cancel, refund, return or replace anything, and it says so.

Managerial implications

A support team does not need an agent that knows everything. It needs one that answers common questions correctly, protects customer data, and passes difficult cases to people with the right context. My results show how each part of the design helps with that.

Capability	Why it matters to a service manager	Evidence from my work
RAG (answers from company documents)	Answers follow the current policy rather than the model's guess. When a policy changes, the team updates a document instead of retraining anything. Each answer can be traced to a source, which helps audits and quality checks.	Decision accuracy rose from 9.6% to 82.7%. Evidence recall on answerable questions rose from 0% to 97.9%.
Automation of routine questions	Delivery, return, refund, warranty and payment questions are repetitive. Handling them automatically can shorten waiting times and leave human agents for complex or sensitive cases.	The agent answered 82.7% of practice questions with the correct decision, and it ran 96 conversation turns with every tool call correct.
Verification before order details	Protects customers and the company from privacy mistakes. Because the model never sees order data before verification, a clever prompt cannot extract it.	Tests confirm blocked details before verification, one retry only, and no verification values in logs.
Consistency	Every caller gets the same policy, in the same tone, at any hour. This reduces the gap between a new and an experienced agent.	Forbidden-claim pass rate was 100% in both the question and conversation tests.
Voice support	Many customers prefer to speak, and voice also helps those who find typing hard. Reusing the same agent means one set of rules to maintain across channels.	Three real recordings produced transcripts, replies and spoken audio through Gnani.
Escalation management	Handover with a ticket reference gives the customer a clear next step and gives staff a record to start from. The agent never promises a refund or replacement it cannot deliver.	Ticket creation is reported only on success, and overdue refunds and high-value damage cases are escalated.

My recommendations for a real rollout

- **Start with a pilot.** Begin with low-risk topics such as delivery, returns and warranty, and keep a person one step away for money and complaint cases.
- **Track the right numbers.** Beyond accuracy, a manager should watch how many cases are escalated, how many customers repeat their question, how long a case takes to resolve, and customer satisfaction.
- **Review failures every week.** I would sample escalations and wrong answers, fix the documents or rules, and re-run the evaluation before each change goes live.
- **Keep documents clean.** The agent is only as good as its policies. Mark old versions as superseded and give each section an owner.
- **Test voice more widely.** Before launch, test with more speakers, accents, noisy conditions and longer calls, and measure speech accuracy and delay.

Reproducing my run

I set the dataset folder and my own LLM and Gnani keys locally, and I never commit the keys. I then run `run.py --check`, the full evaluation with `evaluate.py --top-k 6`, the audio run with `run.py --audio-dir`, and `build_report.py` to produce the report. The source archive excludes the dataset, secrets and virtual environments.